Physics 4C NAME: Lotuscana Ng

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1. Answer the following.

a. Based on the conversions between Celsius and Fahrenheit and between Celsiusand Kelvin, determine absolute zero on the Fahrenheit scale.

$$\begin{cases} T_C = T_K - 273.15 \\ T_F = \frac{9}{5} T_C + 32 \end{cases}$$

$T_{C,0} = T_{K,0} - 273.15$

$T_{C,0} = 0 - 273.15 = -273.15\,^\circ C$

$T_{F,0} = \frac{9}{5} T_{C,0} + 32 = \frac{9}{5}(-273.15\,^\circ C) + 32\,^\circ F$

$= (-491.67 + 32)\,^\circ F$

$= \boxed{-459.67\,^\circ F}$

b. Determine kT for room temperature, where k is Boltzmann's constant (you will need to look up the value). Express the answer in electron-volts.

$k = 1.38 \times 10^{-23}\, m^2\, kg\, \frac{1}{s^2 K}$, room temp = 20°C

$T_{C,rt} = T_{K,rt} - 273.15 \rightarrow T_{K,rt} = (20 + 273.15)\,^\circ C = 293.15\,K$

$kT = \left(1.38 \times 10^{-23}\, m^2\, kg\, \frac{1}{s^2 K}\right)(293.15\,K)$

$= 404.547 \times 10^{-23}\, \frac{m^2 kg}{s^2} = 4.045 \times 10^{-21}\, J = 4.045 \times 10^{-21}\, J \times \frac{1\, ev}{1.602 \times 10^{-19}\, J}$

$= \boxed{2.525 \times 10^{-2}\, ev}$

2. A copper ball ($\alpha = 17 \times 10^{-6}/K$) of radius 13.010cm is placed in a 13.000cm radius circular hole of an aluminum plate ($\alpha = 23 \times 10^{-6}/K$). Both the ball and plate are at 20 °C.

a. To what temperature must the copper ball and aluminum plate both be raised or lowered to in order for the ball to be able to pass through the plate?

Let r_b be the radius of the ball.

$\Delta r_b = \alpha_b r_{b,i} \Delta T$

Let L_p be the circumference of the plate.

$L_p = 2\pi r_p$

$L_{p,i} = 2\pi r_{p,i}$

$\Delta L_p = \alpha_p L_{p,i} \Delta T = \alpha_p 2\pi r_{p,i} \Delta T$

$\Delta L_p = 2\pi \Delta r_p \rightarrow 2\pi \Delta r_p = \alpha_p 2\pi r_{p,i} \Delta T$

$\hookrightarrow \Delta r_p = \alpha_p r_{p,i} \Delta T$

Looking for ΔT s.t.

$13\,cm + \Delta r_p = 13.01\,cm + \Delta r_b$

$13\,cm + \alpha_p r_{p,i} \Delta T = 13.01\,cm + \alpha_b r_{b,i} \Delta T$

$(\alpha_p r_{p,i} - \alpha_b r_{b,i}) \Delta T = 0.01\,cm$

$$\Delta T_K = \frac{0.01\,cm}{\alpha_p r_{p,i} - \alpha_b r_{b,i}} = \frac{0.01\,cm}{(23\times10^{-6}\frac{1}{K})(13.0\,cm) - (17\times10^{-6}\frac{1}{K})(13.01\,cm)}$$

$$= \frac{0.01\,cm}{(299 - 221.17)\times10^{-6}\frac{cm}{K}}$$

$$= \frac{1\times10^{-2}}{77.83\times10^{-6}}\,K = 0.01285\times10^{4}\,K = 128.5\,K$$

$T_{c,f} = T_{c,i} + \Delta T_K = T_{c,i} + \Delta T_C = (20 + 128.5)\,^\circ C = \boxed{148.5\,^\circ C}$

b. Same problem, except consider a 13.010cm *aluminum* ball on top of a 13.000cm circular hole in a *copper* plate.

$$\Delta T_K = \frac{0.01\,cm}{\alpha_b r_{b,i} - \alpha_p r_{p,i}} = \frac{0.01\,cm}{(23\times10^{-6}\frac{1}{K})(13.01\,cm) - (17\times10^{-6}\frac{1}{K})(13.0\,cm)}$$

$$= \frac{0.01\,cm}{(299.23 - 221)\times10^{-6}\frac{cm}{K}} = \frac{1\times10^{-2}}{78.23\times10^{-6}}\,K$$

$$= 0.012783\times10^{4}\,K = 127.83\,K$$

$T_{c,f} = (20 + 127.83)\,^\circ C = \boxed{147.83\,^\circ C}$

c. What if the ball and the plate are both made of copper?

$$\Delta T_K = \frac{0.01\,cm}{(13.01 - 13)(17\times10^{-6})\frac{cm}{K}} = \frac{0.01}{0.01\,(17\times10^{-6})}\,K = 0.05882435\times10^{6}\,K = 588824.35\,K$$

$T_{c,f} = (20 + 588{,}824.35)\,^\circ C = \boxed{608{,}824.35\,^\circ C}$

3. (10 points) 10g of ice at 0 ° C is placed into a 35g aluminum cup filled with 50g of water, both initially at 30 ° C. The mixture is thermally isolated from its environment and allowed to come to equilibrium.

a. (5 points) What is the final temperature of the mixture?

Let Q_i, Q_c, Q_w be the energy absorbed by the ice, cup, and water respectively. Let Q_{latent} be the energy exerted to phase change ice → water.

Then $Q_{latent} + Q_i + Q_c + Q_w = 0J$ by conservation of energy.

$m_i L_w + c_i m_i \Delta T_i + c_c m_c \Delta T_c + c_w m_w \Delta T_w = 0\,J$

$m_i L_w + c_i m_i (T_f - T_{i,i}) + c_c m_c (T_f - T_{c,i}) + c_w m_w (T_f - T_{w,i})$

$= (10g)(334\frac{J}{g}) + (4.18\frac{J}{gC})(10g)(T_f - 0°C) + (0.9\frac{J}{gC})(35g)(T_f - 30°C) + (4.18\frac{J}{gC})(50g)(T_f - 30°C)$

$= 3340\,J + 41.8\frac{J}{C}T_f + 31.5\frac{J}{C}T_f - 945\,J + 209\frac{J}{C}T_f - 6270\,J$

$= -3875J + 282.3\frac{J}{C}T_f = 0J$

$282.3\frac{J}{C}T_f = 3875J$

$T_f = \frac{3875}{282.3}C = \boxed{13.72°C}$

b. (5 points) How much ice (at 0 ° C) should be added to bring the final temperature of the mixture down to 0 ° C?

$L_f m_{ice} = 7215J$

$m_{ice} = \frac{7215}{333}g = 21.7g$

$21.7g - 10g = \boxed{11.7g}$